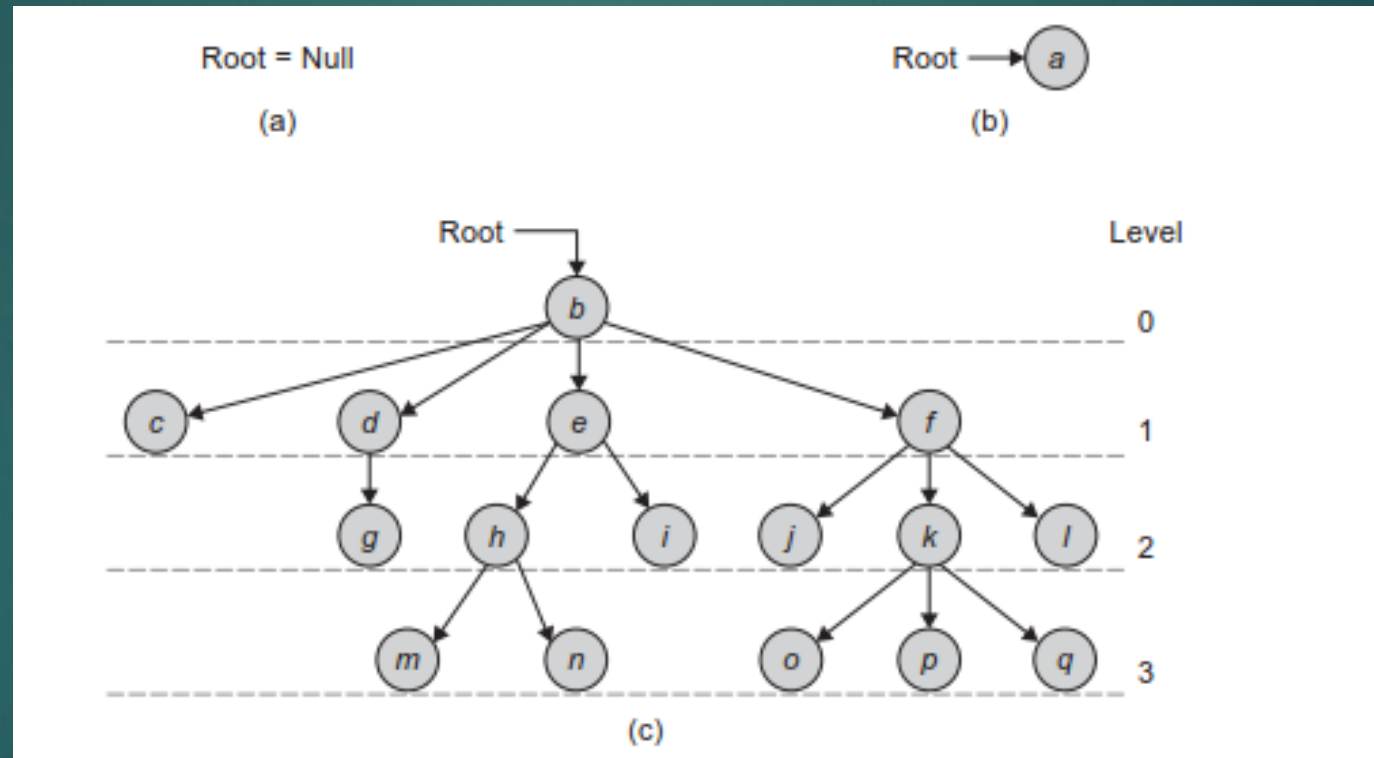


Basic Terminology: Trees

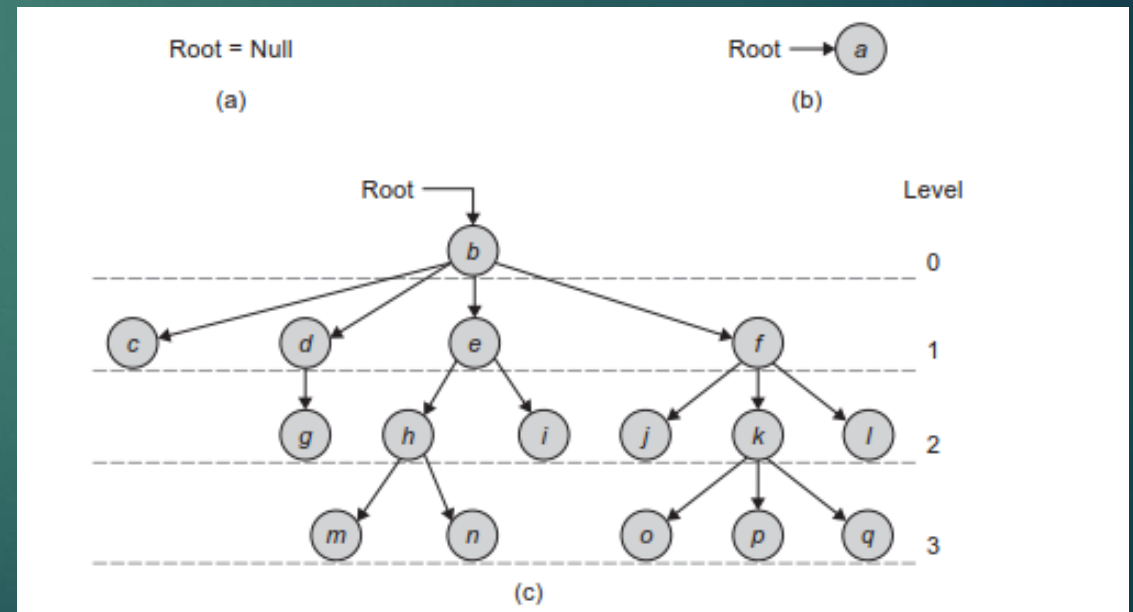
- ▶ **Directed tree** An acyclic directed graph is a directed tree.
- ▶ **Root** A directed tree has one node called its root, it's the topmost element. It's a node that doesn't have any parent of its own.
- ▶ **Parent node:** the immediate predecessor of any node, root is not having any parent node.
- ▶ **Child node:** immediate successor of a node
- ▶ **Grand Parent**
- ▶ **Terminal node (leaf node)** In a directed tree, any node that has an outdegree zero/ no child node is a terminal node. The terminal node is also called as leaf node (or external node).
- ▶ **Branch node (internal node)** All other nodes whose outdegree is not zero are called as branch nodes.

Basic Terminology: Trees



Basic Terminology: Trees

- ▶ A directed path (or simply path) is a sequence of directed edges $e_1, e_2, \dots, e_n$, from source node to destination.
- ▶ $b \rightarrow e \rightarrow h \rightarrow m$



Basic Terminology: Trees

- **Ancestor**- any predecessor node from the root to that node.

Ancestor of n- **b, e, h**

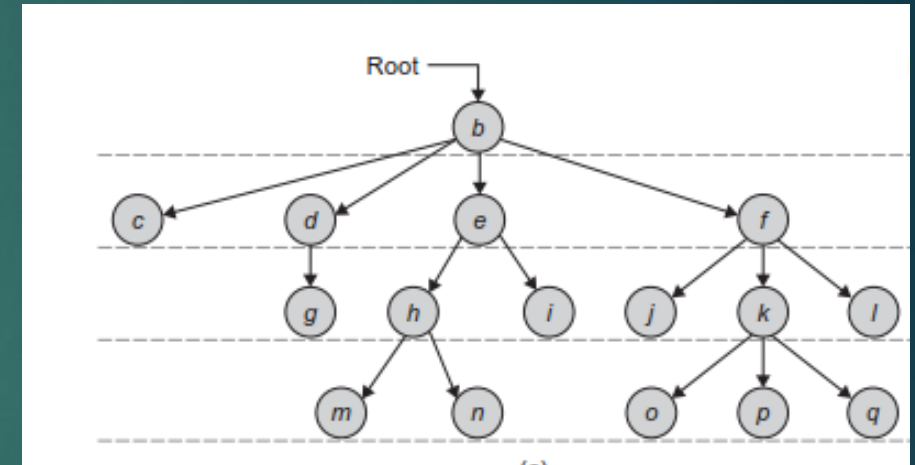
Ancestor of o- ?

Descendant – Any successor node from a node to the leaf node

Descendant of e= h,l,m,n

Common Ancestor of h and j?

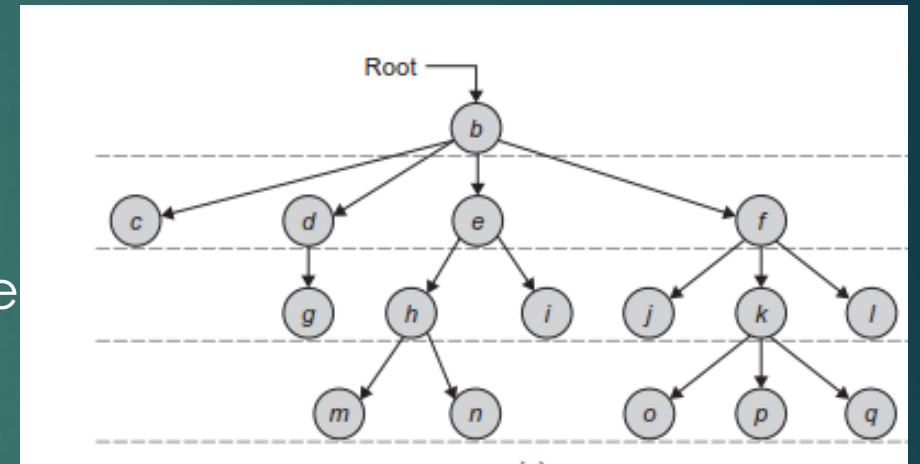
Common descendant of h and j?



Basic Terminology: Trees

- ▶ **Subtree**: containing a node with all its descendants
- ▶ **Sibling**: All the children of same parent
- ▶ **Degree of a node**- All the child nodes of a node
- ▶ **Degree of a tree**- maximum degree of a node

Here the degree of tree is 4



Basic Terminology: Trees

- **Depth of a node**- length of a path from root to that node.

depth of h: 2

depth of j: ?

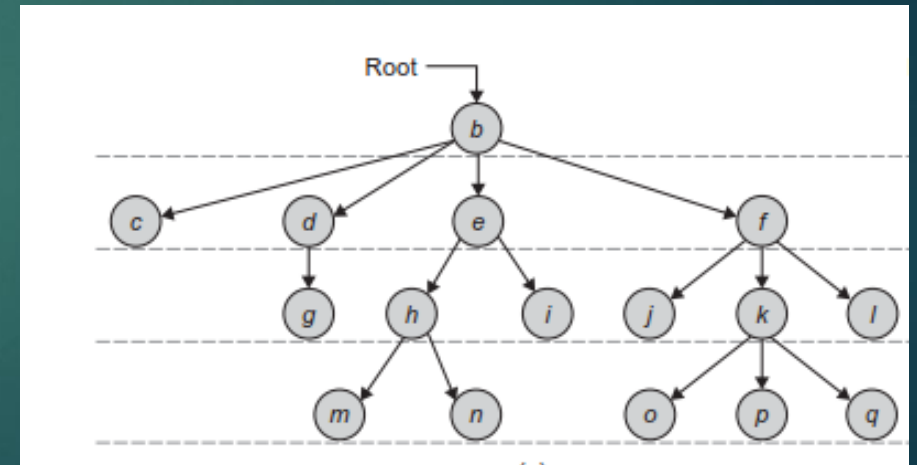
depth of a: ?

Height of node: number of edges in the longest

Path from that node to the leaf node

Height of node e= 2

Height of tree=?



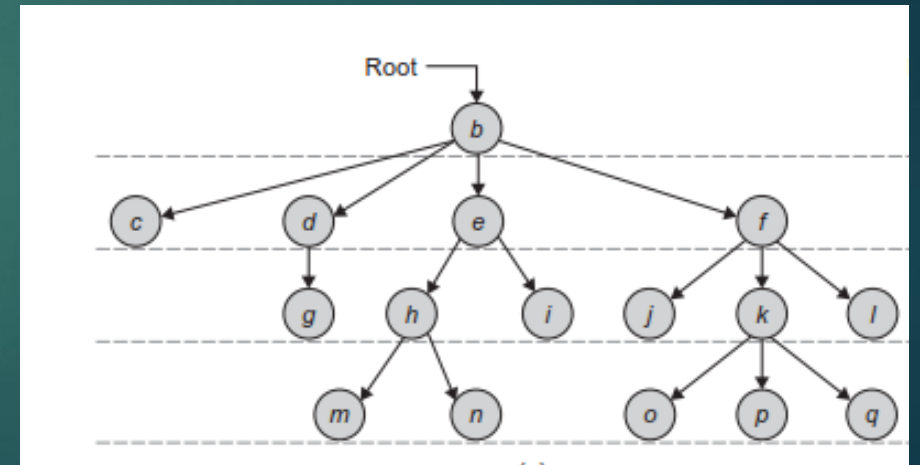
Basic Terminology: Trees

- ▶ **Level of node**: distance from root to the given node

level of b = 0

Level of g = 2

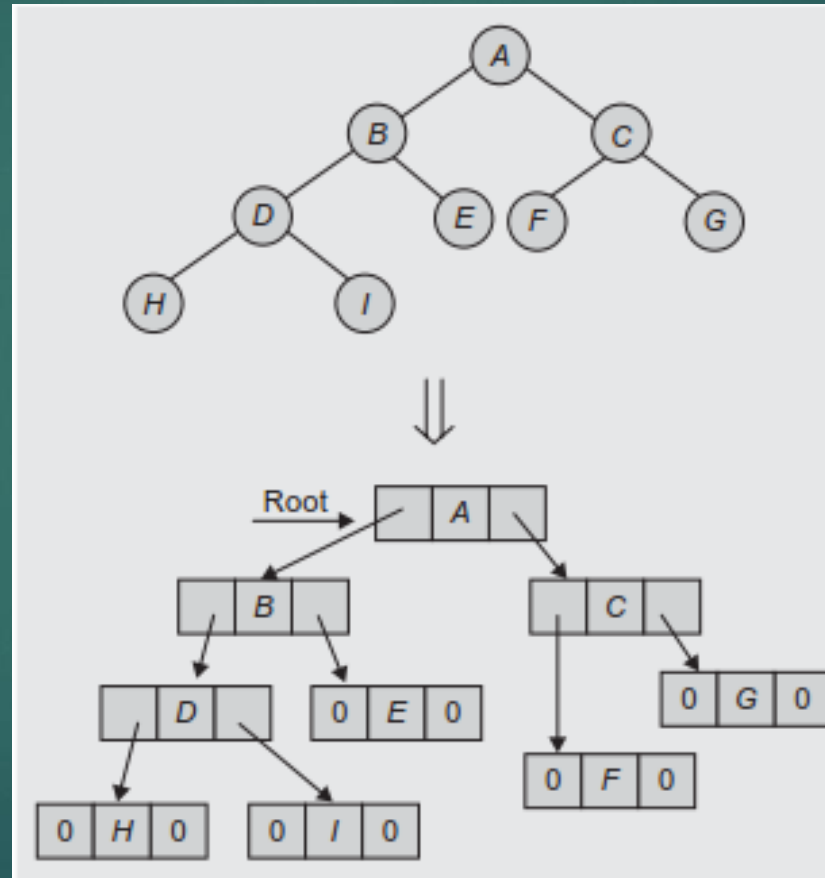
- ▶ Level of node is always equal to depth of node
- ▶ Level of tree is equal to height of tree
- ▶ Level of a node is not equal to height of node



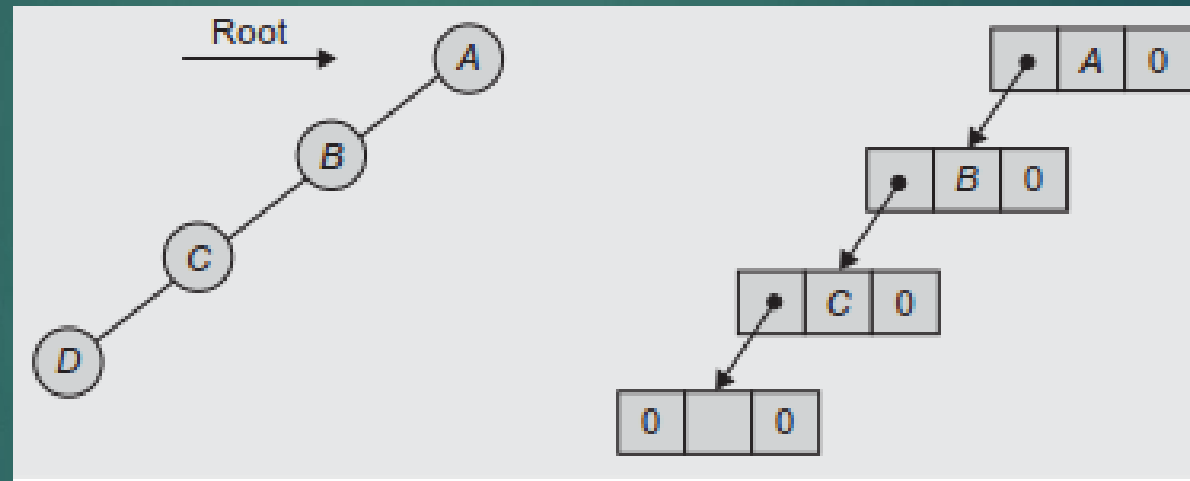
Basic Terminology: Trees

- ▶ If a tree has n nodes then number of edges are $n-1$

Representation of tree

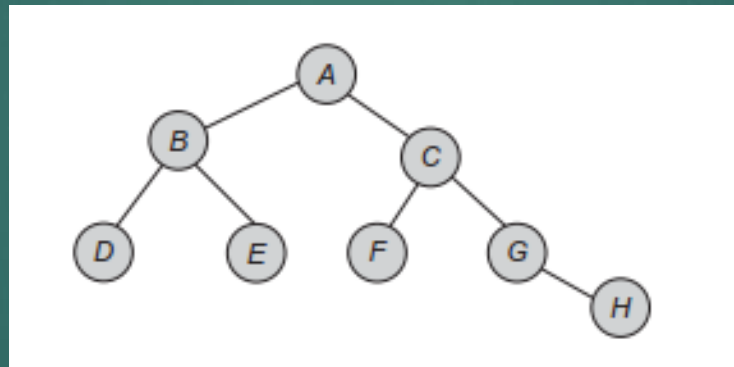


Representation of tree

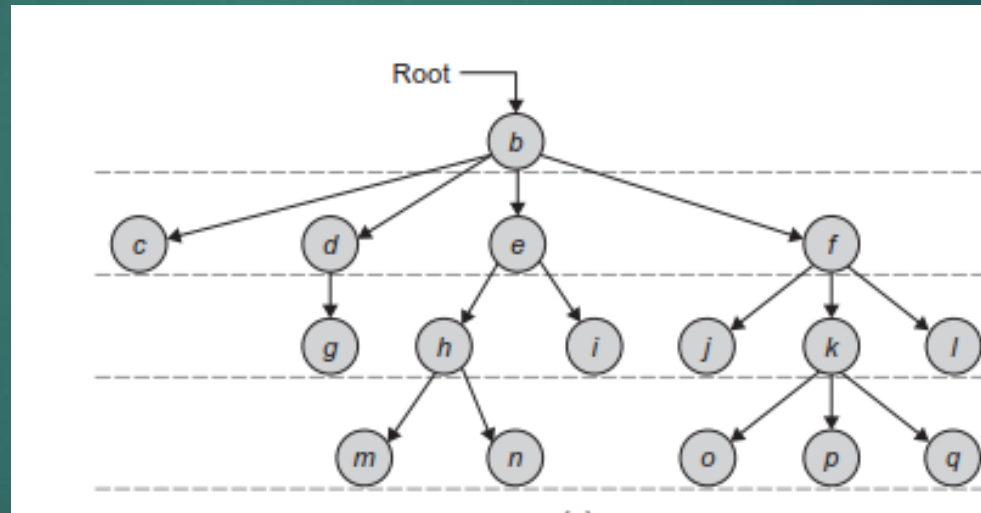


Binary Tree

- ▶ Each node can have at most 2 child nodes.
- ▶ Either 0, 1 or 2



Binary Tree

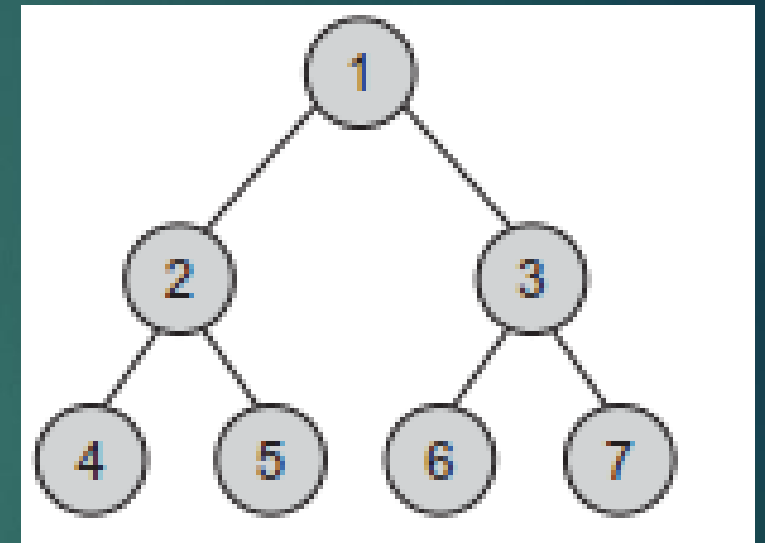


Binary Tree

- ▶ At level 0- max nodes possible 1
- ▶ At level 1- max nodes possible 2
- ▶ At level 2- max nodes possible 4
- ▶ At level 3- max nodes possible 8

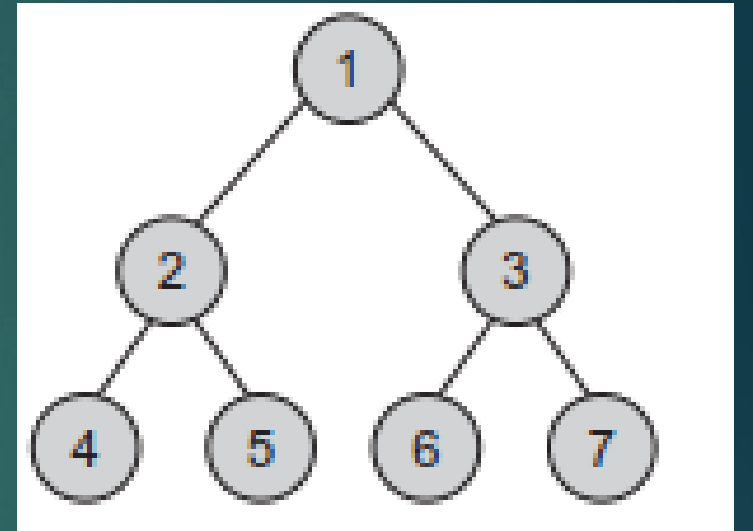
- ▶ $2^0 + 2^1 + 2^2 + 2^3$

- ▶ At each level i = maximum number of nodes possible is 2^i



Binary Tree

- ▶ Maximum number of nodes possible at height h
- ▶ Height of root node=height of tree
- ▶ At height 3, maximum number of nodes possible :
 $1+2+4+8=15$
- ▶ If height is 2, maximum number of nodes possible :
 $1+2+4=7$
- ▶ Maximum number of nodes possible at height $h=2^{h+1}-1$



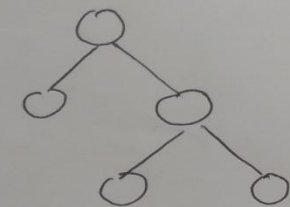
Types of Binary Trees

- ▶ Full/proper/strict
- ▶ Complete binary tree
- ▶ Perfect binary tree
- ▶ Degenerate binary tree

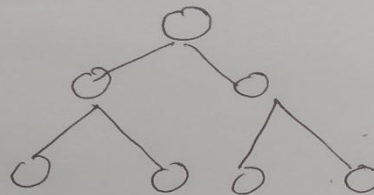
Full Binary Tree

- ▶ **Full binary tree**: each node will contain exactly 2 nodes except the leaf node.
- ▶ No. of leaf nodes = no. of internal nodes + 1

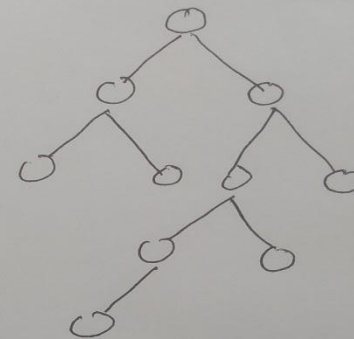
Full binary tree



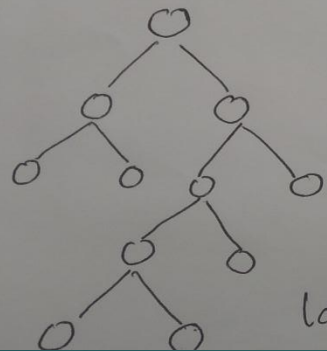
(a)



(b)



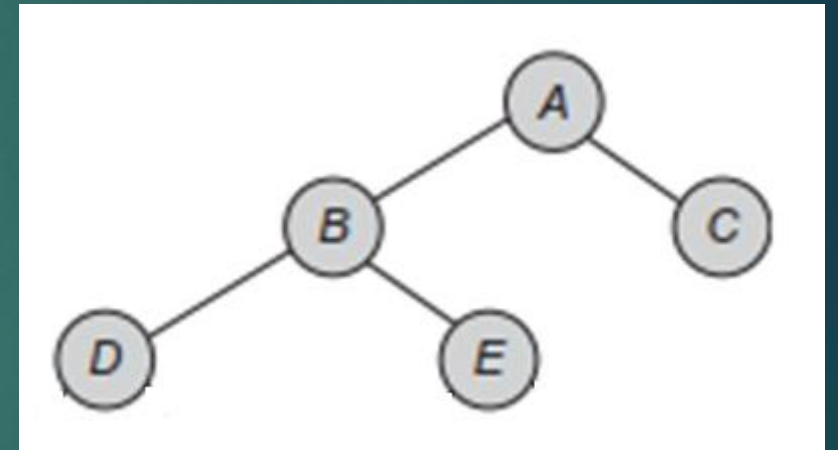
(c)



(d)

Full Binary Tree

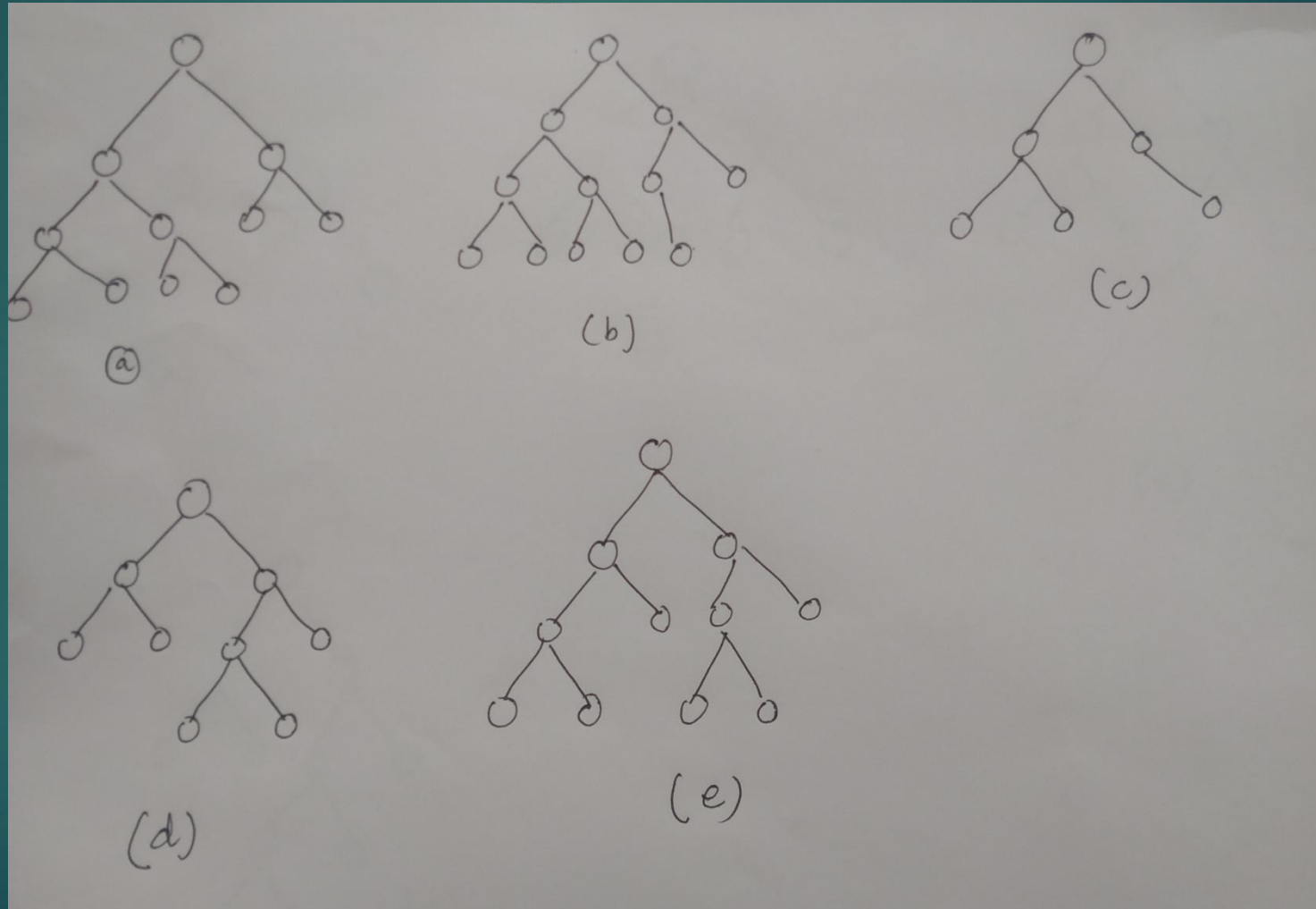
- ▶ Minimum number of nodes in Full binary tree
- ▶ At height 0= 1 node
- ▶ At height 1= 2 node
- ▶ At height 2= 5 nodes
- ▶ At height 3= 7 nodes
- ▶ Minimum number of nodes= $2h+1$



Complete binary tree

- ▶ Complete binary tree:
- ▶ all the levels are completely filled (except possibly the last level) and the last level has nodes as left as possible.

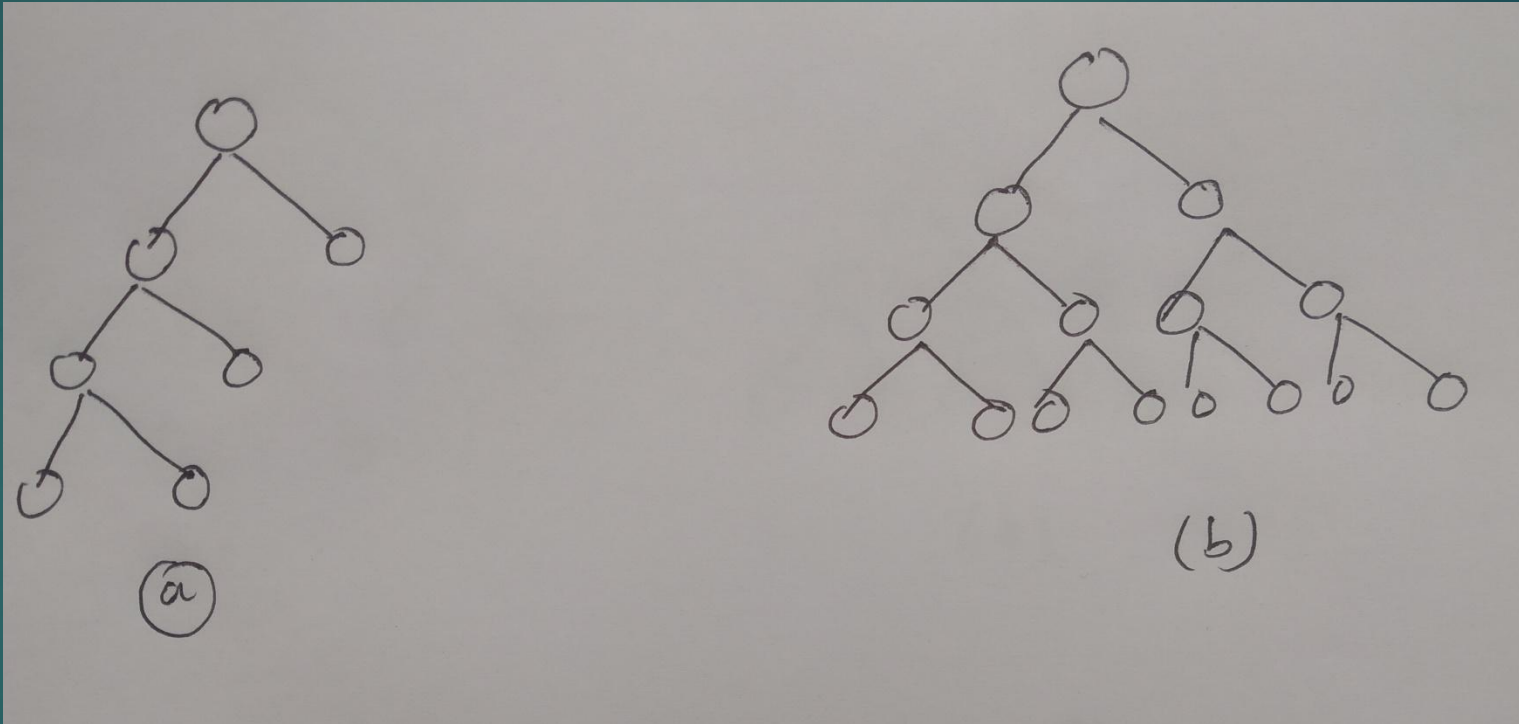
Complete binary tree



Perfect binary tree

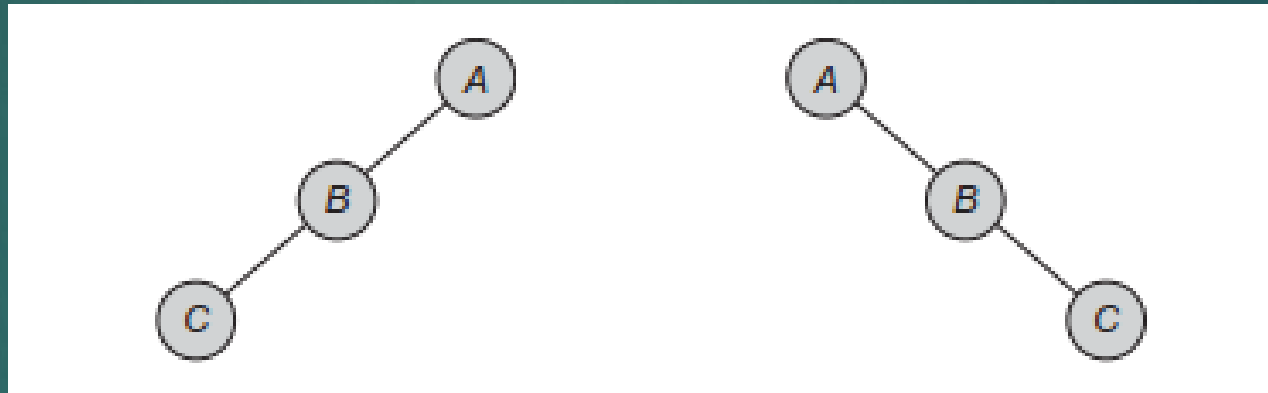
- ▶ Perfect binary tree
- ▶ If all the internal nodes are having 2 children and all the leaf nodes are at the same level.

Perfect binary tree



Degenerate binary tree

- ▶ Degenerate binary tree:
- ▶ All the internal nodes have just one child.



Left skewed binary tree
Right skewed binary tree